Separating Proof from Judgement in Reference-Free Evaluation of Japanese News Summaries

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50 XL-Sum Japanese articles, 250 candidate summaries

***Abstract*—A summariser in production is handed an article and returns a summary; nobody hands it a human reference. This report describes an evaluator built to that contract. Exploration showed the 250 supplied candidates are not a quality gradient but five distinct populations, and that the two most dangerous—fluent text that reverses the article, and fluent text that invents its central claim—are invisible to every cheap signal, because they are on topic by construction. The design separates constraints a script can prove from constraints only a reader can judge, and requires an independent reviewer to confirm every stopping decision. Validation rests on seven checks that do not share a failure mode, including a third independent read of the 109 candidates for which no ground truth can be derived. The evaluator ships as an installable plugin, implemented twice on two agent hosts; the two agree on routing for 94.4 % of candidates and on terminal category for 74 of 74.**

***Index Terms*—LLM evaluation, reference-free metrics, summarisation quality, agent isolation, auditability, reproducibility**

I. Exploration

A. First contact: what the 250 candidates actually are

The dataset is 50 articles of 483–3664 characters, each with a human reference of 41–164 and five candidates of 22–302.

Reading candidates side by side inside their articles, rather than as a flat list, made the structure obvious within the first few articles: the five candidates per article are not five samples from one generator. They are draws from distinct populations. A string-level scan of the corpus alone—no model, no scoring—confirmed and counted them (Table I): four are mechanically derivable, the fifth is the remainder.

TABLE I
Corpus census.

Population	n	Derivation
Reference reproduced	58	candidate = reference
Copy of the article	50	candidate ⊂ article body
Reference fragment	17	reference starts with candidate
Belongs elsewhere	16	matches a *different* article
Generated, unlabelled	109	none of the above
Total	**250**	

Note. Categories derived from the corpus by string comparison alone, with no evaluator output involved; used as weak labels only after both runs were final.

Three structural facts came out of the same scan. *The sentence limit is almost never the problem*—only three of 250 candidates exceed three sentences, and all three are article copies that would be stopped anyway, so a rule enforcing the product specification is necessary but will almost never fire. *The corpus contains an accidental controlled experiment*: twenty groups of candidates are byte-identical after Unicode normalisation, eight inside a single article and fourteen spanning two—a free reliability instrument.

The references are not ground truth, and the data says so. The assignment warns that XL-Sum reference quality is uneven. It is worse than uneven: several references assert facts the article body never contains. One reference states a 4 割 sexual-assault share, a suspect profile and a 379-case figure, none of which appear in the supplied text; another is a photo caption opening with この写真, a demonstrative whose antecedent exists only in an image nobody is given. Any design scoring a candidate by its distance to the reference rewards it.

TABLE II
Hypotheses formed during exploration and their fate.

Hypothesis	Outcome
Copying is detectable with strings, not embeddings	**Survived.** A copy is maximally similar to its article, so similarity is the wrong instrument; containment and n-gram coverage caught 50 of 50.
Embedding similarity separates topic mismatch	**Survived.** Two 10-article probes: 0.707–0.868 matched against 0.163–0.398.
Similarity can screen relevance on its own	**Weakened.** On all 250 pairs the bands overlap, off-topic reaching 0.4813 and on-topic falling to 0.4034: it may nominate, not decide.
Missing sentence-final punctuation indicates truncation	**Killed.** 体言止め, a verbal noun closing a headline, is standard news register and looks identical to an amputated predicate.
Factual defects form one gradient, so one faithfulness score suffices	**Killed.** A wrong peripheral number and a reversed main event are not two points on a scale but a flawed summary and a false one.

B. Hypotheses, and which ones died

Five hypotheses were formed before building anything. Two survived unchanged, one survived in weakened form, and two were killed by the data (Table II).

The last row is the finding the design turns on. Inspecting the 109 unlabelled candidates one by one exposed a failure mode with no detector anywhere in the pipeline as it then stood: text that is fluent, within the sentence limit, not a copy, not truncated, on topic, and *states the opposite of the article*. One article reports the Japanese prime minister announcing a dissolution of the lower house; its candidate reports him announcing he will *not* dissolve it. Nothing upstream can see this: the string evidence is clean and the embedding similarity *high*, because the topic is right—which is precisely why the failure is dangerous. A sibling mode invents rather than inverts, attributing to the US president a line, 「米国はもはや世界

の警察ではない」, that appears nowhere in the article and displaces the headline's actual claim.

Set against these, one candidate is a different animal: it substitutes five details (シリア for イラク, 男性 for 女性, 30 代 for 10 代, 8000 for 6000, 500 for 300) while the central event, a fierce IS-clearing battle in Mosul's old city, is correctly reported. It is a bad summary, not a false one, and collapsing both into "score zero" throws the ranking away.

C. Contributions

These observations yield one design, implemented twice on two agent hosts. The contributions are as follows.

1) *Data characterisation.* The 250 candidates form five distinct populations rather than a quality gradient, only 141 of them labellable, and the references are not ground truth. The two dominant failure modes, reversal and fabrication, are on topic by construction, beyond any cheap signal.
2) *A stated determinism–judgement boundary.* Hard constraints terminate and the rest is graded on a soft rubric; inside the hard tier a script decides only what its measurement proves while a Grounding Gate agent screens grounding before any scoring, which settles 51 of 250 pairs deterministically and spares the most expensive stage 80 more.
3) *Judge as a plugin.* Where LLM-as-a-judge is normally a notebook and a prompt, we deliver it as an installable plugin whose role contracts, rubric and few-shot bank are versioned files and whose records carry a replayable trace, so a third party can run and audit the evaluator rather than rebuild it—and, being portable, implement it twice, turning delivery itself into a validation instrument.
4) *Validation by instruments that fail differently.* The two implementations agree with corpus-derived weak labels at 0.915 and with each other on routing for 94.4 % of candidates; on the 109 candidates carrying no derivable label a third independent reading agrees with the submitted run at $\rho = 0.960$, and every number here is recomputable.

II. Design

A. The evidence boundary

The runtime evaluator sees exactly one article and one candidate—never a reference, never another article, never a corpus—because a caller who sent one request has nothing else to give. An earlier gate ranked each candidate against every article in the batch; it worked, and was removed anyway, because it measured something the deployed system cannot. References enter once, after every score is final (Section III).

B. Two layers of hard constraint, then a graded score

Figure 1 is the funnel, annotated with the actual counts from the submitted run. The organising principle is that constraints divide by what kind of evidence can settle them.

Layer 1 is physical. Empty output, sentence count above three and near-verbatim copying are settled by a script, because there the measurement *is* the conclusion: a containment ratio near 1.0 is not evidence of copying, it is copying.

TABLE III
Soft rubric.

Dimension	Max	Question	Mean
Faithfulness	50	Is every claim supported by the article?	38.7
Coverage	30	Is the main event and its key facts present?	19.0
Coherence	15	Complete, grammatical, ordered?	13.5
Conciseness	5	Compact, without repetition?	4.8
Total	**100**		**76.1** over 165 survivors

Note. Scored only for candidates surviving both hard layers; the four dimensions sum to the total exactly. **Mean** is run A (Claude Code plugin) over its 165 graded survivors.

Layer 2 is semantic. Whether a candidate is grounded cannot be settled by any measurement, so a Grounding Gate agent screens every survivor before any scoring, returning at most one central category (OFF_TOPIC, FACTUAL_REVERSAL, FABRICATED_CONTENT) with an article span and a candidate span as evidence. It is deliberately blind to quality—no rubric, no anchor, no score—because an agent that both screens and scores trades a central defect for a low score.

Between the two layers sits embedding relevance, *required but never terminal*. Every survivor must leave that stage carrying a similarity value against its own article—a run that could not reach the embedding model is incomplete, not cheaper—but a value below threshold is only a hint that the semantic layer should look closely.

Survivors of both layers receive a graded score (Table III). Faithfulness carries half the total because fluent-but-false is the dominant risk here; conciseness carries five points because the sentence limit is already a hard constraint.

Terminal outcomes all score 0 but keep separate rank tiers, so five candidates remain orderable even when several fail. Reversal, fabrication and off-topic share the worst tier, because each misleads a reader who cannot check the source; copying and truncation rank above them.

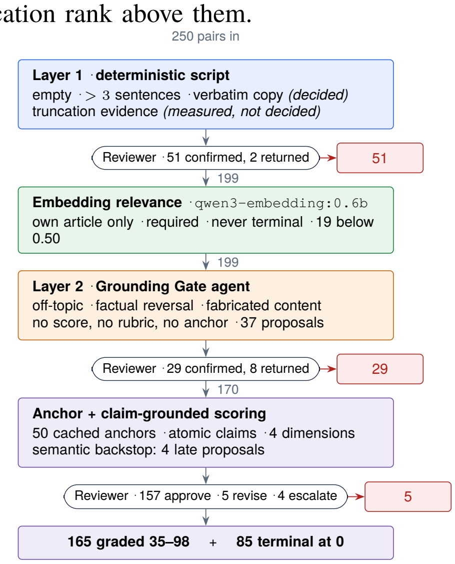

Fig. 1. The funnel, annotated with counts from the submitted 250-pair run. Red exits are Reviewer-confirmed terminations. The 51 pairs stopped at layer 1 cost one script pass and one confirmation; 80 of 250 never reach anchor generation and scoring, the most expensive stage.

C. Independent review of every stopping decision

No stage may stop a candidate on its own authority: every proposal goes to a Reviewer in a fresh context that re-derives the evidence from the article. The review is not a rubber stamp—it rejected 2 of 53 physical proposals, 8 of 37 grounding proposals (including all eight fabrication claims) and 3 of 4 late scorer proposals, and escalated four truncations the gate had not proposed. Disagreement surviving one revision round is preserved as low confidence rather than averaged away.

D. Alternatives considered and rejected

Table IV lists the approaches tried and dropped, each with the measurement that killed it. Two are worth naming. A cross-encoder reranker (`bge-reranker-v2-m3`) separates off-topic text more cleanly in mean terms but buys no recall at five times the false alarms. And embedding the candidate against the generated anchor correlated better with coverage yet predicted it worse under leave-one-out—a feature that correlates better and predicts worse is fitted to the sample, so the anchor stayed a reasoning aid and never became a number.

TABLE IV
Rejected approaches and the evidence that rejected them.

Approach	**Why it was rejected**
Score against the reference	Unavailable in production, and wrong here: generated candidates beat 45 of 50 references.
Similarity as a quality score	A verbatim copy is the most similar text there is.
Cross-encoder reranker	Same recall (16/16), five times the false positives (15 vs. 3 of 234) at a 0.50 cut.
Anchor similarity as a scoring feature	Higher correlation, worse leave-one-out error.
Batch-relative relevance	Uses articles a production caller never sent; removing it changed 0 of 19 nominations.
Regex decides truncation	体言止め is indistinguishable from an amputated predicate without the deleted text.
One agent screens *and* scores	It trades a central defect for a low score instead of stopping.

E. Delivery: the design is a plugin, not a script

The evaluator is packaged as an installable plugin, so anyone who needs it can run it rather than reconstruct it. The primary implementation—**run A** below—is a Claude Code plugin carrying four isolated agents (Grounding Gate, Scorer, Reviewer, Reporter) and the deterministic scripts they call. There is no orchestration notebook and no hidden state: the rubric, the few-shot bank and each agent's contract are files under version control, so a reader can see what every role was told. The same design is implemented again as a Codex plugin, **run B**, which is the instrument for the cross-host check in Section III-E. Both emit the same record schema, including a trace of every stage visited with its evidence, so any score replays backwards to the measurement that produced it.

III. Validation

An evaluation is worth only its weakest justification, so the goal was several checks that fail differently rather than one repeated. Table V states what the two implementations produced; Tables VI–IX give each check's numbers.

Naming used throughout. Run A is the Claude Code plugin, the primary implementation and the source of the submitted scores; **run B is the Codex plugin**, an independent second implementation on a different host with a different underlying model, used as cross-host validation, never as a fallback. Every table with an A and a B column repeats it.

A. Run-level results

Table V is the whole outcome of both runs; A is the submitted one. They land on the *same* count for the failure a script can prove—50 verbatim copies—and within one for the failure an embedding can nominate, 16 against 17 off-topic; they diverge only on the two categories a reading must raise, A stopping 13 reversals and 1 fabrication against B's 7 and 0. Their distributions differ in shape, not centre: B has the wider spread and awards nearly twice as many top-band scores, so it is the more generous reader while A stops eight more candidates. B's orchestration also sent far more drafts back for a second round, a fact about the hosts rather than the design; both runs end with all 250 records approved and nothing escalated. Every disagreement below follows from it.

TABLE V
Run-level results.

	A	**B**		**A**	**B**
Pairs / articles	250/50	250/50	*Dimension mean*		
Routing			Faithfulness /50	38.7	37.4
Terminal	85	77	Coverage /30	19.0	20.0
Graded	165	173	Coherence /15	13.5	13.9
Terminal category			Conciseness /5	4.8	4.8
VERBATIM_SOURCE_COPY	50	50	*Quality label*		
OFF_TOPIC	16	17	EXCELLENT (90–100)	28	52
FACTUAL_REVERSAL	13	7	GOOD (75–89)	66	49
OBVIOUS_TRUNCATION	5	3	FINE (65–74)	34	26
FABRICATED_CONTENT	1	0	MIXED (50–64)	25	26
Graded score			POOR (0–49)	12	20
Mean / median	76.1/79	76.2/82	*Review completion*		
SD	15.1	19.0	Final decision APPROVE	250	250
Q1 / Q3	68/87	63/91	Unresolved escalations	0	0
Min / max	35/98	28/100	Settled in one round	242	95
Record integrity			Sent back for a second	8	155
Dimension-sum errors	0	0	Reviewer confidence HIGH	196	250
Label-band violations	0	0			

Note. **A** = Claude Code plugin (primary run, the source of the submitted scores); **B** = Codex plugin (secondary). *Terminal*: stopped by a confirmed hard-constraint decision, scored 0. *Graded*: reached the soft rubric.

Across all 500 records, every soft score equals the sum of its four dimensions and every label falls inside its band—the last two rows of Table V. This is a floor, not a result.

B. Determinism and a natural experiment in pair conditioning

The corpus supplies two free instruments (Section I-A). Eight groups of candidates are byte-identical *and* attached to the same article, each scored independently without either scorer knowing the twin existed. All eight received identical totals and identical dimension values in both implementations, so run-to-run variance explains none of the spread below.

Fourteen further groups, 32 candidates, are byte-identical but split across articles. If the evaluator graded text quality both copies would score the same; they do not. In all 14, in both implementations, the foreign placement was terminated at 0 as off-topic and the native placement graded on its merits—one such text scores 87 under its own article and 0 under another—the evaluator conditions on the pair, which a text-only metric cannot.

C. Agreement with corpus-derived weak labels

After both runs were final, references were opened for the first time and the Table I categories were used as weak labels. Table VI and Fig. 2 give the per-population outcome.

TABLE VI
OUTCOME BY CORPUS-DERIVED POPULATION.

		Stop		Survivor mean	
Population	n	**A**	**B**	**A**	**B**
Copy of the article	50	50	50	—	—
Belongs elsewhere	16	16	16	—	—
Reference fragment	17	5	3	59.4	48.4
Reference reproduced	58	0	0	78.6	79.9
Generated	109	14	8	76.6	77.9
Agreement, 141 labelled		129 / 127		0.915 / 0.901	

Note. **A** = Claude Code plugin (primary run, the source of the submitted scores); **B** = Codex plugin (secondary). *Stop*: candidates terminated; the score columns cover survivors only. *Agreement*: routing matches the population's expectation—copies, foreign placements and fragments stop, reference reproductions do not.

Survivor ranges tell more than the means: fragments survive at 39–70 and 34–63, reference reproductions at 42–89 and 40–99, generated candidates at 35–98 and 28–100, and no surviving fragment reaches GOOD. Two things in Fig. 2 matter more than the headline figure. The fragment population is graded rather than uniformly stopped—a deliberate divergence from the label, discussed in Section IV-B—and in run A *no* reference reproduction reached EXCELLENT while 28 generated candidates did, the reference-quality problem of Section I made quantitative. Run B puts 10 references in the top band against 42 generated candidates, so the ordering survives.

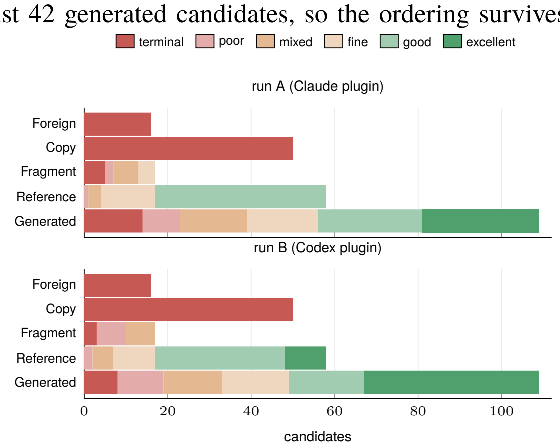

Fig. 2. Outcome by corpus-derived population, both runs. The two populations that should never survive never do, in either implementation; the fragment population is graded rather than stopped; and the reference population sits below the top of the generated population in both.

D. Within-article ordering

The application needs candidates ordered inside an article, so ordering was checked directly rather than inferred from scores (Table VII). Across all 50 articles, in both runs, zero articles placed a planted failure at or above the reference reproduction, and the ordering is not degenerate: every article retained two to four survivors with a wide spread between them, and the reference reproduction sits at rank 2 in the large majority and never below rank 3. It is consistently near the top without being treated as the ceiling, which is what a design that distrusts its references should produce.

TABLE VII
WITHIN-ARTICLE BEHAVIOUR OVER THE 50 ARTICLES.

	A	B		A	B
Planted failure $\geq$ ref.	0	0	*Survivors per article*		
Rank of the reference reproduction			2 survivors	4	3
rank 1	5	7	3 survivors	27	21
rank 2	43	36	4 survivors	19	26
rank 3	2	7	0, 1 or 5	0	0
rank 4 or 5	0	0	*Score spread among survivors*		
			median	26	32
			min / max	6 / 61	2 / 68

Note. **A** = Claude Code plugin (primary run, the source of the submitted scores); **B** = Codex plugin (secondary). *Survivors*: candidates that reached the soft rubric. *Spread*: best minus worst survivor within one article.

E. Two implementations, two hosts

The strongest available check on a design—as opposed to a run—is to implement it twice and compare. The Codex plugin was written against the same rubric but a different host and a different underlying model, and neither implementation saw the other's output. Table VIII is the comparison; Fig. 3 is the same data as a scatter and a difference distribution.

TABLE VIII
CROSS-IMPLEMENTATION AGREEMENT.

	count	rate		count	rate
Routing, $n = 250$			*Score,* $n = 162$		
Agreement (both ways)	236	0.944	Spearman ρ		0.891
both terminal	74		Pearson r		0.865
both graded	162		Mean $\lvert\Delta\rvert$		6.85
terminal in A only	11		Mean Δ (SD)		−1.97 (8.81)
terminal in B only	3		$\lvert\Delta\rvert \leq 5$	86	0.531
Terminal-category	74 / 74	1.000	$\lvert\Delta\rvert \leq 10$	124	0.765
Per dim., $n = 162$*:* ρ */ mean* $\lvert\Delta\rvert$			$\lvert\Delta\rvert \leq 15$	145	0.895
Faithfulness /50	0.864 / 4.25		$\lvert\Delta\rvert \leq 20$	156	0.963
Coverage /30	0.882 / 2.96		Label identical	88	0.543
Coherence /15	0.635 / 0.70		Label within one band	157	0.969
Conciseness /5	0.243 / 0.25				

Note. **A** = Claude Code plugin (primary run, the source of the submitted scores); **B** = Codex plugin (secondary). Routing rows cover all 250 pairs, score rows the 162 both implementations graded. Δ = A − B; ρ is Spearman, r is Pearson.

The disagreements are one-sided and more informative than the agreements. All 11 pairs stopped only by A are severity calls on a defect B also found—6 reversals, 4 fragments and 1 fabrication, which B scored between 28 and 66—and the 3 stopped only by B were scored 58–71 by A. Not one is a defect that only one implementation saw: the two disagree about where the cliff is, not which candidates are near it, and all 14 come from the populations needing a judgement.

The per-dimension block needs a caveat. Faithfulness and coverage, which carry 80 of the 100 points and do the discriminating, correlate well. Coherence and conciseness correlate far worse, yet differ by a fraction of a point, because both runs push almost every survivor to the ceiling—range restriction, not disagreement, and a sign that 15 and 5 points are more than these dimensions earn. Exact agreement on the five-band label is only 88 of 162, the least flattering number in the report; it is a banding artefact, since 96.9 % of pairs fall within one band,

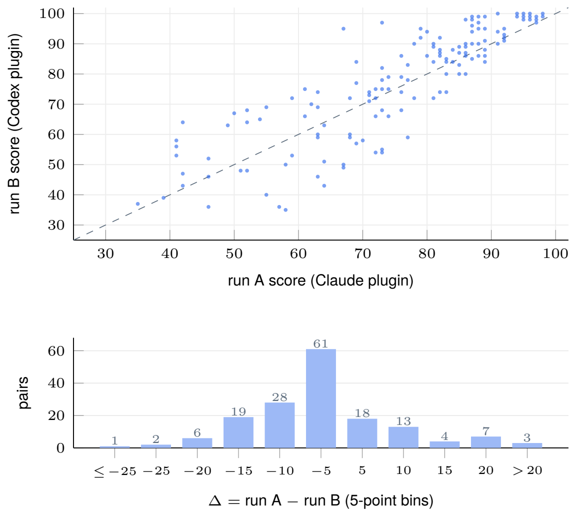

Fig. 3. The 162 candidates graded by both implementations. $\rho = 0.891$, mean $|\Delta| = 6.85$. Above: the dashed line is equality. Below: the signed difference is centred slightly below zero (-1.97), the long tail is A scoring lower, and 96.3 % of pairs fall within 20 points.

but the bands are a convenience for a reader and the ordering is what should be trusted.

F. The block with no ground truth

Every check so far rests on the 141 labellable candidates, the easy ones. The 109 generated candidates—44 % of the corpus, and the block carrying the fluent factual errors—have no derivable label, so reporting 0.915 while staying silent about them reports the wrong number. They were scored a third time by a separate reading with no access to either run's output (Table IX): on the 94 that all three readings graded it agrees with the primary run at $\rho = 0.960$ and the secondary at $\rho = 0.936$, both closer than the two implementations agree. Every quantity quoted in this report is re-derived by a checker that reads only the corpus and the run outputs, and that fails if any of its 199 assertions moves.

TABLE IX
Third independent reading of the 109 unlabelled candidates.

Pair	n	ρ	r	**mean** $\lvert\Delta\rvert$
Third read vs. A	94	0.960	0.951	4.09
Third read vs. B	94	0.936	0.924	5.95
A vs. B	94	0.916	—	6.52
Group	n	**third-read score**		
Terminated by A	14	mean 32.5, range 20–53		
Terminated by B	8	mean 34.2, range 22–59		
Neither	95	mean 77.0		

Note. **A** = Claude Code plugin (primary run, the source of the submitted scores); **B** = Codex plugin (secondary). Upper block: the 94 candidates all three readings graded. Lower block: candidates one run terminated, scored by a reader not told so.

The lower block of Table IX is the part that matters. A reader who was never told which candidates had been stopped placed every one of them at the bottom of the distribution —mean 32.5 for the 14 that run A terminated and 34.2 for B's 8, against 77.0 for the rest—so the terminal decisions on the unlabelled block are corroborated by a source that did not make them. This does not make the scores correct, since three readings of the same rubric can share a bias and none is a human judgement, but the signal there is reproducible.

IV. Limitations

A. What the evaluation does not measure

No human judgement is involved, so agreement reflects reproducibility rather than correctness, and the weak-label check can itself accept a flawed reference. Beyond grammatical completeness the evaluation does not assess style, register, audience fit or newsworthiness. It asks only whether the facts a candidate selects are the article's central ones.

B. What it measures poorly

Truncation is the weak point, and structurally so: a reference cut from 「…方針を発表した。」to 「…方針を発表」was judged untruncated because 発表 is a verbal noun discharging the clause, and only the deleted した。distinguishes 体言止め from an amputated predicate. The fragment population therefore splits: mid-token endings are visible and missing them is a recall problem worth fixing, while 体言止め endings are undecidable under a reference-free contract, so the 5-of-17 figure overstates the deficiency. The five-band labels are also less stable than the scores (Section III-E), and cost is uneven, since moving the truncation judgement to the Reviewer makes a fragment consume both stages before it is stopped.

C. One structural gap, stated plainly

The Reviewer could technically access the corpus and its reference column, although no reference was provided or recorded as used. This is a procedural safeguard, not structural evidence. The next step is to re-adjudicate the 85 terminal cases with the reference-blind Reviewer and report the resulting delta.

D. What I would do next

More data, above everything else: the report rests on 250 pairs from 50 articles of one publisher in one language, so every threshold here—the 0.50 similarity cut, the dimension maxima, the band boundaries—should be assumed invalid outside that sample. The immediate work is the same funnel over a larger Japanese corpus from several publishers, then across languages, since constraints built on Japanese sentence segmentation and 体言止め register are language-specific and must be re-derived from scratch. Only then is human adjudication of a stratified sample, and controlled perturbation of numbers, entities, negation and causality, worth the spend.

V. Conclusion

The design rests on one claim: each constraint is decided by the evidence that can settle it, and no stage stops a candidate on its own authority. On this corpus it held—no ordering inversion against a reference, 0.915 against the weak labels, 94.4 % routing agreement with an independent twin, and $\rho = 0.960$ from a third reading. And because it installs as a plugin with every score carrying its trace, the next corpus is a run, not a rebuild.